Energetic Constraints in Spatial World Models

Reinterpreting Fei-Fei Li's Spatial Intelligence through the Energetic Paradigm

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Abstract

This article proposes a reinterpretation of AI world models through the Energetic Paradigm. While world models represent objects and spatial relations within an environment, they do not specify how those states should evolve during reasoning and planning. As a result, systems often rely on uncontrolled trajectory search in state space. The Energetic Paradigm reframes the world model as an energy system in which states correspond to energy configurations and transitions correspond to energy redistribution. Under this view, reasoning becomes the evolution of energy configurations rather than arbitrary trajectory exploration. This structure organizes how spatial environments evolve during AI reasoning and reduces uncontrolled exploration in world models.

A. The Real Difficulty of Intelligence

Artificial intelligence is often described today as a system that is becoming increasingly powerful through scale. Larger models, larger datasets, and larger computing infrastructure have allowed AI systems to generate language, images, and even video with impressive realism. Many observers therefore believe that intelligence will simply emerge once systems become large enough. However, the real difficulty of intelligence does not lie in producing text or images. The real difficulty lies in understanding and acting within the structure of the physical world.

Consider a simple environment such as a kitchen. A human being entering the room instantly understands the relationships between objects. A cup resting near the edge of a table might fall if pushed slightly forward. A chair placed behind the table blocks a walking path. A refrigerator door cannot open if another object stands too close. These judgments are made almost instantly and rarely require conscious calculation. Humans appear to possess a deep and stable understanding of spatial relationships and environmental constraints.

Current AI systems, despite their remarkable abilities in language generation and pattern recognition, frequently fail in such situations. A model may generate a sophisticated explanation of architecture or physics while simultaneously failing to predict that an object might fall from a table edge. The system may correctly identify objects but remain uncertain about how they interact in a physical environment. This contrast reveals that modern AI systems excel at representation but remain limited in their ability to reason within spatial structures.

The Spatial Gap

Fei-Fei Li has emphasized that this limitation reflects a deeper structural gap in artificial intelligence. Early breakthroughs in deep learning were driven primarily by the organization of large datasets. The ImageNet project reorganized visual data on an unprecedented scale and enabled neural networks to learn visual categories effectively. This transformation demonstrated that structural changes in data organization could unlock new capabilities in machine learning systems.

However, recognizing objects is only one part of intelligence. Recognizing that an image contains a table, a cup, and a chair does not necessarily imply understanding how those objects occupy space or how they interact with one another. Real-world intelligence requires systems that can understand spatial relationships, physical constraints, and environmental dynamics. This broader capacity has become known as spatial intelligence.

Spatial intelligence introduces the concept of a world model. A world model is an internal representation of the environment in which objects, positions, and relationships form a structured system. Within such a system, an AI agent can simulate possible future situations and evaluate the consequences of actions before executing them. In principle, this allows the system to anticipate events and plan behavior in complex environments.

B. The Problem of Expanding State Space

Although world models represent an important step toward more realistic artificial intelligence, they introduce a new difficulty. Once a system attempts to represent the entire environment as a set of possible states, the number of possible configurations grows extremely rapidly. Even a simple environment can generate an enormous number of potential states as objects move, interact, and change positions.

In computational terms, the environment can be represented as a state space. Each state corresponds to a particular configuration of the environment, including the positions of objects, their orientations, and their possible motions. When an agent

performs an action, the system transitions from one state to another. Planning therefore involves evaluating sequences of state transitions that form possible trajectories through the environment.

The difficulty arises because the number of possible trajectories grows exponentially with the number of steps considered. A system that attempts to evaluate every possible future quickly encounters a combinatorial explosion. The computational cost of exploring the entire state space becomes prohibitive, and the system may generate many unrealistic or physically inconsistent states during its exploration.

For example, an unconstrained world model might generate hypothetical states in which objects overlap, move through solid surfaces, or appear in physically impossible positions. Although such states can later be filtered out, the system has already spent computational resources exploring them. As environments become larger and more complex, this inefficiency becomes increasingly problematic.

Trajectory Explosion

This problem can be understood as trajectory explosion. When the system attempts to predict the future, it generates many potential trajectories that represent different ways the world might evolve. Each trajectory is a sequence of states linked by possible actions or environmental events. Without structural guidance, the system must examine a large number of trajectories before identifying those that are feasible or meaningful.

Many modern AI approaches address this problem through optimization techniques such as reinforcement learning or policy search. These methods attempt to learn which trajectories lead to desirable outcomes. However, they still rely on extensive exploration of the state space during training or planning. As a result, the computational burden remains high, and the system may still encounter unstable or unrealistic trajectories.

The central question therefore becomes how to organize the exploration of state space more effectively. Instead of searching blindly through all possibilities, an intelligent system should ideally restrict its exploration to trajectories that are structurally plausible within the environment.

C. Reconstructing Systems Through Energy

The Energetic Paradigm proposes a different way of thinking about this problem. Instead of treating the world as a collection of arbitrary states, the paradigm interprets the system as an energy structure. In this view, the environment is not merely a set of

configurations but a system in which relationships between components carry energetic significance.

Within such a system, each configuration of the environment can be associated with a certain energetic condition. Transitions between configurations involve changes in this energetic structure. Rather than exploring all possible transitions, the system evaluates whether a transition is energetically feasible within the structure of the environment.

This perspective shifts the focus of reasoning. Instead of searching through a vast state space for possible trajectories, the system examines how energy can be redistributed within the environment. Certain transitions naturally follow from the existing energy configuration, while others require large energetic changes that make them unlikely or unstable.

Mathematically, a trajectory through the environment can be described as a sequence of states

$$\tau = (s_0, s_1, s_2, \ldots, s_T)$$

Each transition between states can be associated with an energetic cost

$$E(s_i \rightarrow s_{i+1})$$

The total energetic cost of a trajectory becomes

$$E(\tau) = \sum_{i=0}^{T-1} E(s_i \rightarrow s_{i+1})$$

Only trajectories whose total energetic cost remains within feasible limits are considered valid.

Energy-Constrained Evolution

This formulation does not simply introduce an arbitrary constraint. Instead, it reorganizes the reasoning process itself. The system no longer explores all possible states equally. Instead, it moves through the environment along paths that preserve structural coherence in the energetic system.

For example, a cup resting on a table occupies a stable configuration in which gravitational forces and surface contact maintain equilibrium. A transition in which the cup moves gradually across the table requires relatively little energetic change. By contrast, a transition in which the cup suddenly appears floating in midair would require a large energetic shift that violates the underlying structure of the system.

When reasoning within this framework, the AI system naturally favors transitions that maintain environmental stability. Unrealistic or physically inconsistent states carry high energetic cost and therefore become unlikely paths of exploration. As a result, the system avoids spending computational resources on trajectories that cannot occur within the structure of the world.

In this way, the Energetic Paradigm does not merely add a constraint to an existing algorithm. It reinterprets the environment as an energy system in which reasoning corresponds to the redistribution of energetic relationships. Planning becomes the process of finding feasible energetic transformations rather than searching blindly through an enormous state space.

D. Energetic Evaluation of States and Transitions

The decisive question is not whether a world model should be constrained, but how such constraint is formally constructed. A useful answer cannot stop at slogans such as low-energy path or feasible transition. It must define what energy means, how energy is represented, how energy structure is built from a spatial world model, and how that structure changes the reasoning process itself. The goal of this section is therefore to move from intuition to mechanism. Once this mechanism is stated clearly, the Energetic Paradigm is no longer a decorative label placed on top of planning, but a redefinition of how spatial reasoning is organized.

A spatial world model begins with a set of entities, their attributes, and their relations inside an environment. Let the world state at time $t$be denoted by

$$s_t = \{o_1, o_2, \dots, o_n, r_{12}, r_{13}, \dots, r_{ij}, c_t\},$$

where o_i represents an object, r_{ij} represents a relation between objects i and j, and $c_t$denotes contextual variables such as support conditions, terrain, or task-relevant environmental context. In a conventional world model, this state is treated as a representational description of the environment. The model then predicts the next state through a transition function

$$s_{t+1} = F(s_t, a_t),$$

where a_t denotes an action or a system event. The problem is that this representation by itself does not distinguish structurally plausible transitions from implausible ones. It describes possibilities, but it does not organize them.

The Energetic Paradigm begins at precisely this point. It does not replace the world model. It transforms the world model from a descriptive structure into an organized energetic field. That transformation begins by defining an energy function over states:

$$E(s_t): \mathcal{S} \to \mathbb{R}_{\geq 0}.$$

Here, energy is not identical with physical energy in the narrow sense of mechanics. It is a structural scalar measuring how difficult, unstable, incoherent, or implausible a state is within the logic of the represented environment. A state with low energy is not merely possible. It is structurally coherent. A state with high energy is not merely unlikely. It violates the organization of the environment.

From State Description to Energy Field

The crucial move is that energy must be derived from the internal structure of the represented world rather than assigned as an arbitrary penalty. For that reason, the energy of a state should be decomposed into interpretable components. A general form is

$$E(s_t) = \lambda_g E_g(s_t) + \lambda_p E_p(s_t) + \lambda_t E_t(s_t) + \lambda_r E_r(s_t),$$

where E_g is geometric energy, E_p is physical energy, $E_t$is temporal energy, and E_r is relational energy. The coefficients $\lambda_g, \lambda_p, \lambda_t, \lambda_r$ determine the relative contribution of each component. This formula matters because it prevents the term energy from becoming vague rhetoric. Once the decomposition is written down, each part can be defined, computed, learned, and tested.

Geometric energy measures the degree to which the spatial arrangement of objects respects the geometry of the environment. If two objects occupy incompatible positions, geometric energy rises. If an object extends beyond a surface that should support it, geometric energy rises. If a body appears discontinuously inside a blocked region, geometric energy rises. One possible form is

$$E_g(s_t) = \sum_{i<j} \phi_{\text{overlap}}(o_i, o_j) + \sum_i \phi_{\text{boundary}}(o_i) + \sum_i \phi_{\text{support}}(o_i),$$

where ϕ_{overlap} penalizes impossible intersection, ϕ_{boundary} penalizes violations of environmental boundaries, and ϕ_{support} penalizes unsupported placements. This is the first place where the Energetic Paradigm becomes operational. Geometry is no longer passive description. It becomes a source of energetic structure.

Physical energy measures whether the state respects the causal and dynamical logic of the environment. In a simple setting this may include gravity, balance, friction, contact, inertia, or containment. In a richer setting it may include articulated motion, deformable interaction, or fluid constraints. A general expression can be written as

$$E_p(s_t) = \sum_i \psi_{\text{stability}}(o_i) + \sum_{i<j} \psi_{\text{contact}}(o_i, o_j) + \sum_i \psi_{\text{dynamics}}(o_i).$$

If a cup rests stably on a table, the stability term remains low. If the same cup appears floating without support, the stability term rises sharply. If two rigid objects occupy mutually exclusive positions, the contact term rises. If velocity or acceleration violates the physical logic already present in the scene, the dynamics term rises. In this formulation, physical plausibility is converted into a measurable field over state space.

Temporal energy addresses a failure that standard world models often hide behind representational adequacy. A single frame may look plausible even when the transition that produced it is not. The system therefore requires an explicit cost on temporal discontinuity. This can be written as

$$E_t(s_t, s_{t-1}) = \sum_i \left(\alpha \parallel x_i^{(t)} - x_i^{(t-1)} \parallel^2 + \beta \parallel v_i^{(t)} - v_i^{(t-1)} \parallel^2 + \gamma \parallel q_i^{(t)} - q_i^{(t-1)} \parallel^2 \right),$$

where $x_i^{(t)}$ is position, $v_i^{(t)}$ is velocity, and $q_i^{(t)}$ is orientation or pose of object $i$at time t. The constants α, β, γweight spatial displacement, motion change, and pose change. This term ensures that a sequence of individually plausible frames does not conceal an implausible transition. Temporal energy turns continuity into a formal requirement rather than an intuitive afterthought.

Relational energy captures the fact that a spatial world is not reducible to object coordinates. Objects stand in structured relations: inside, above, supporting, blocking, contained by, reachable from, occluding, aligned with, and so on. These relations often determine whether a world state is coherent at all. A useful relational term is

$$E_r(s_t) = \sum_{(i,j)\in\mathcal{R}} \omega_{ij}\ \chi_{ij}(r_{ij}^{(t)}, \hat{r}_{ij}^{(t)}),$$

where $r_{ij}^{(t)}$ is the observed or inferred relation between objects $i$and j, $\hat{r}_{ij}^{(t)}$ is the structurally expected relation, ω_{ij} is an importance weight, and χ_{ij}measures relational inconsistency. If a chair is represented as simultaneously behind and intersecting a table, relational energy rises. If an object is marked as contained within a closed cabinet while visually remaining outside, relational energy rises. The environment is thus organized not only by coordinates, but by relational coherence.

From Energy over States to Energy over Transitions

Once energy is defined over states, the next step is to define the energetic structure of transitions. The Energetic Paradigm does not merely ask whether a state is plausible. It

asks whether the movement from one state to the next follows a structurally coherent path. A transition cost can therefore be written as

$$\Delta E_t = E(s_{t+1}) - E(s_t),$$

but this alone is not enough, because a transition may preserve total state energy while still being dynamically implausible. The proper transition functional must include both state energy and transition-specific cost:

$$C(s_t \to s_{t+1}) = E(s_{t+1}) + \mu T(s_t, s_{t+1}),$$

where $T(s_t, s_{t+1})$ is a transition penalty and $\mu$is a scaling coefficient. A useful expansion of T is

$$T(s_t, s_{t+1}) = \eta_g T_g + \eta_p T_p + \eta_t T_t + \eta_r T_r,$$

with geometric, physical, temporal, and relational transition penalties corresponding to the state-level structure defined above. This formula matters because it distinguishes a low-energy destination from a coherent path to that destination. A system can no longer teleport to an apparently plausible state without paying transition cost.

A trajectory is then evaluated not as a loose sequence of predicted futures, but as a cumulative energetic process. Let a trajectory be

$$\tau = (s_0, s_1, \dots, s_T).$$

Its total energetic cost is

$$\mathcal{E}(\tau) = E(s_0) + \sum_{t=0}^{T-1} C(s_t \to s_{t+1}).$$

This is where the real methodological difference appears. Standard planning searches over trajectories. The Energetic Paradigm organizes state space into an energy landscape and then evaluates trajectories as energetic flows across that landscape. The question is no longer simply which future state is rewarding, but which sequence of transformations preserves structural coherence at acceptable energetic cost.

E. Constructing the Energetic Structure

The next question is methodological rather than conceptual. How is this energy structure obtained in practice. There are three routes, and a mature system may combine all three.

The first route is structural derivation from the world model itself. If the world model already contains object geometry, support relations, physical attributes, occlusion, contact, and temporal tracking, then many energetic terms can be computed directly. In that case, energy is not externally imposed. It is derived from the represented environment. If a support graph exists, unsupported states can be penalized. If collision volumes exist, overlap penalties can be computed. If temporal trajectories exist, discontinuity costs can be estimated. This route is analytically clean because the energy structure grows directly from the structure already present in the model.

The second route is learning from observed trajectories. Suppose the system observes a dataset of real spatial evolutions

$$\mathcal{D} = \{\tau^{(1)}, \tau^{(2)}, \dots, \tau^{(N)}\}.$$

The energy function can then be learned so that observed trajectories receive low energy and implausible or corrupted trajectories receive high energy. One objective is

$$\min_{\theta} \sum_{\tau \in \mathcal{D}} \mathcal{E}_{\theta}(\tau) - \sum_{\tilde{\tau} \sim q} \mathcal{E}_{\theta}(\tilde{\tau}),$$

where θ parameterizes the energy model and $\tilde{\tau} \sim q$ denotes corrupted, sampled, or adversarial trajectories. This converts the environment into an energy-based learning problem. Plausible spatial evolution is not merely labeled as good. It is embedded into the energy field itself.

The third route is latent energetic construction. Many world models operate not directly in explicit object space, but in a latent representation space. Let the latent world state be z_t, produced by an encoder

$$z_t = \Phi(s_t).$$

Then the energetic structure can be defined in latent space as

$$E_z(z_t) = \rho(z_t) + \kappa(z_t, z_{t-1}) + \sigma(z_t, \mathcal{C}),$$

where ρmeasures intrinsic latent plausibility, κmeasures temporal continuity in latent evolution, and σ measures compatibility with contextual constraints $\mathcal{C}$. This route is especially important when object-level symbolic structure is incomplete or unavailable. In that case, energy still organizes the world model, but the organization is learned over latent geometry rather than hand-specified entities.

F. What the System Then Does

Once the energy structure has been built, the reasoning process changes in a precise way. The world model first proposes candidate next states, either through dynamics prediction, simulation, policy rollout, or latent decoding. The energetic layer then evaluates each candidate through the state and transition function. Candidate states are not all passed equally into planning. They are ranked, filtered, or reweighted by energetic coherence. Formally, candidate selection can be written as

$$\mathcal{S}_{t+1}^{*} = \left\{ s' \in \hat{\mathcal{S}}_{t+1} \mid C(s_t \to s') \leq \epsilon_t \right\},$$

where $\hat{\mathcal{S}}_{t+1}$ is the set of generated candidate states and ϵ_t is an adaptive energetic threshold. The system thus does not reason over all generated futures. It reasons over the energetically admissible subset.

A stronger version does not merely filter candidate states after generation. It uses energy to reshape generation itself. The transition model becomes

$$p(s_{t+1} \mid s_t, a_t) \propto \exp\left(-C(s_t \to s_{t+1})\right),$$

so that the probability of generation is directly modulated by energetic structure. This is methodologically important. It means the Energetic Paradigm is no longer post hoc correction. It becomes part of the generative dynamics of the world model itself. The model learns not only to represent the world, but to evolve it along structurally coherent gradients.

At the level of planning, trajectory selection becomes

$$\tau^{*} = \arg\min_{\tau \in \mathcal{T}} \mathcal{E}(\tau),$$

possibly combined with task reward:

$$\tau^{*} = \arg\min_{\tau \in \mathcal{T}} \left[\mathcal{E}(\tau) - \beta R(\tau)\right].$$

This equation shows where Energetic Paradigm and standard control or reinforcement learning can be joined without collapsing into each other. Reward still encodes task preference. Energy encodes structural coherence. The system therefore does not simply pursue reward. It pursues reward along trajectories that remain compatible with the energetic organization of the environment.

Why This Is Not Mere Constraint Decoration

This section must end by making the methodological point absolutely clear. If energy is added only as an afterthought, then the system has merely acquired another penalty term. But if the state space is reconstructed as an energy field, if transitions are evaluated as energetic flows, if candidate generation is reweighted by energetic structure, and if planning is performed over energetic trajectories rather than arbitrary sequences, then the system has changed its reasoning ontology. It is no longer moving through a neutral map of possible states. It is moving through a structured landscape whose geometry determines which transformations are easy, coherent, expensive, unstable, or forbidden.

That is the actual method. Energy is defined as a scalar field over spatial states and transitions. The field is decomposed into geometric, physical, temporal, and relational terms. It is built from explicit structure, learned trajectories, or latent organization. It is integrated into candidate generation, transition evaluation, and long-horizon planning. The result is not simply that the world model is constrained. The result is that the world model is reorganized into an energetic system, and reasoning becomes the controlled evolution of that system.

G. Implications for Future Intelligence

If artificial intelligence systems adopt an energy-based interpretation of world models, several important consequences follow. First, reasoning becomes structurally grounded in the environment rather than purely computational. The system's exploration of possible futures reflects the energetic relationships present in the world itself.

Second, the effective size of the search space is dramatically reduced. Because the system only considers energetically feasible transitions, large portions of the state space never need to be explored. This reduction can significantly improve the efficiency of planning and simulation in complex environments.

Third, the integration of spatial intelligence and energetic reasoning provides a more coherent architecture for artificial intelligence. World models supply the representation of objects, positions, and relationships within the environment. Energetic constraints organize how these states evolve over time. Decision-making algorithms then operate within this structured space to identify feasible actions.

Toward Stable Artificial Systems

Ultimately, the goal of artificial intelligence is not merely to produce impressive outputs but to operate reliably within complex environments. Systems that interact with the

physical world must maintain consistency with the underlying structure of that world. By interpreting environments as energy systems and constraining transitions accordingly, the Energetic Paradigm offers a framework for achieving this stability.

Rather than allowing models to wander through an unrestricted landscape of hypothetical states, the paradigm guides reasoning along paths that remain compatible with environmental structure. In doing so, it transforms the problem of intelligence from a search through arbitrary possibilities into the management of structured energetic transformations.

Such an approach suggests that future progress in artificial intelligence may depend not only on larger models or larger datasets but also on deeper structural understanding of how systems evolve within the environments they represent.